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Population Synthesis Using Incomplete Microsample

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Abstract

This paper presents a population synthesis model based on the Wasserstein Generative-Adversarial Network with Gradient Penalty (WGAN-GP) for training on incomplete microsamples. The proposed method aims to address the challenge of missing information in microsamples on one or more attributes due to privacy concerns or data collection constraints. By using a mask matrix to represent missing values, the study proposes a WGAN-GP training algorithm that lets the model learn from a training dataset that has some missing information. The paper compares the ability of WGAN-GP models trained on incomplete microsamples to those trained on complete microsamples to create a synthetic population. We conducted a series of evaluations of the proposed method using a Swedish national travel survey. We validate the efficacy of the proposed method by generating synthetic populations from all the models and comparing them to the actual population dataset. The results from the experiments showed that the proposed methodology successfully generates synthetic data that closely resembles a model trained with complete data as well as the actual population. The paper makes a contribution to the field by giving a strong solution for population synthesis using incomplete microsamples. It also opens up new research areas and shows how deep generative models can be used to improve population synthesis.

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1. Introduction

Transportation simulation models, which use agent-based models (ABMs), are widely used for predicting travel demand, evaluating policy impacts, or analyzing travel behavior. These models typically require complete information about individuals' social and household characteristics from an area encompassing cities, towns, or even countries (Bastarianto et al. (2023)). Ideally, we could collect such information from individual or household census data and then draw a specific number of samples to create a synthetic population. Many countries' statistical authorities have also made available microsamples of individual-level data from the entire population, which can serve as a substitute for census data. Travel surveys that capture complete demographic and socioeconomic attributes can also act as a

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good replacement. In addition to these microsamples, aggregated marginal information on a regional or zonal level is usually available from the Bureau of Statistics. However, acquiring granular, individual-level data is challenging. Privacy concerns, as well as the technical and financial constraints of data gathering, often impede access to detailed information.

To address data collection challenges, researchers developed population synthesis algorithms to create synthetic populations. They provide ABM transport models with a reliable alternative to actual population data. Population synthesis techniques generate a comprehensive list of synthetic populations, each accompanied by corresponding attribute data. The objective of population synthesis is to optimally utilize the existing microsamples, along with the additional aggregated marginal information on each attribute of interest, in order to generate agents that closely align with the underlying population structure (Sun and Erath (2015)). These simulated agents can thereafter be used to evaluate governmental policies in the region or conduct studies that would be prohibitively costly, ethically questionable, or just unfeasible using actual population data.

As highlighted in the research by Rich (2018) and Borysov et al. (2019), population synthesis methods typically consist of three steps: 1). Creating a starting solution that generates a synthetic pool of individuals to represent various attribute combinations. This is usually done based on the available microsamples. 2). The fitting stage involves estimating the weighting factors for the synthetic pools to create a representative synthetic population for future targets. 3). The allocation stage involves the generation and assignment of synthetic agents to ABM transport models. This paper exclusively concentrates on the initial stage, which involves creating suitable representative samples for a specific population, without taking into account how these samples will align with future objectives. This step is critical because any underlying error in it will propagate into the ABM models, affecting the forecast results.

In the population synthesis literature, there has been a trend toward simulation-based probabilistic models for the first stage. Research from Sun and Erath (2015); Borysov et al. (2019); Garrido et al. (2020); Kim and Bansal (2023) utilized simulation-based generative models to create synthetic populations for the first stage of population synthesis. One crucial similarity in each of these studies is that the microsamples used in the experiments are from a single data source and are complete, i.e., there is no missing information for any attributes in the microsample. The quality of the generated synthetic populations depends, of course, on the quality and detail of the microsample. While the data quality has been improving, it has not kept pace with the growing interest in micro-simulations at the scale of individuals tagged with many associated attributes. Available microsamples are often thin and incomplete. A survey microsample cannot comprehensively cover all the variations of different attributes found in the actual population. Frequently, these microsamples exhibit incomplete data on one or more attributes due to errors in data collection, or the respondents intentionally withholding information or not collecting it to ensure privacy. Additionally, to improve the attribute richness of the microsample, it can be combined with supplemental data from other sources. For example, a travel survey in a certain region from multiple distinct organizations can be combined to add more data and attributes to the microsample. However, there may be situations where one or more attributes are absent from either of the surveys. Hence, there is a need for an imputation algorithm that can be used to estimate missing values based on data that was observed or measured in a microsample. This will ensure that the synthetic population that population synthesis models generate is complete.

In this context, we propose a new approach that draws synthetic populations from incomplete microsamples. In the context of this study, an *incomplete microsamples* is one that has missing information on one or more attributes of the sample. We employ a population synthesis model that utilizes the WGAN as proposed by Kim and Bansal (2023). This study's contribution lies in proposing a novel technique in the WGAN training algorithm that enables the model to effectively learn using incomplete training data. The aim of this study is to propose a methodology that can effectively address the challenges associated with missing data while maintaining a degree of accuracy that is at least equivalent to the methods described in the existing literature.

The remainder of the paper is structured in the following manner: Section 2 presents the proposed training methodology for the WGAN model. We first briefly present the original WGAN training method and then describe the proposed changes to it. To evaluate and assess the performance of the proposed training method, we used a microsample from a Swedish national travel survey. Section refsec:CaseStudy provides the experimental setup, metric evaluation, results, and discussions on these aspects. Finally, in Section 5, the article concludes by summarizing the analysis.

2. Methodology

This paper primarily focuses on the WGAN with Gradient Penalty (WGAN-GP), a generative model based on the framework that generates a synthetic population by transforming randomly generated numbers from a K-dimensional standard normal latent variable, Z. K are the attributes related to the agent's individual characteristics and household characteristics. The generator network, $G(Z)$, is initiated for training with a draw from the latent variable Z. The draw is then transformed in such a way that the output has the same dimensions and shape as the real data. The second network, the discriminator network $D(X)$, receives an observation. This can either be from real data or from the generator $G(Z)$. The discriminator's objective is to determine whether the information it receives is real or not. The training process continues until the D network is no longer able to distinguish between real and generated data. The WGAN-GP aims to generate synthetic population $\hat{P}(X)$ such that they are consistent with actual population, $P(X)$. We refer to study by Gulrajani et al. (2017) and Kim and Bansal (2023) for more details on WGAN-GP model and training.

We used Neves et al. (2022) masking approach to handle missing data. We introduce a new matrix, known as the mask, as input during WGAN-GP training. The mask, Y, represents the training data, X, with a zero representing a missing value and a one representing any non-missing value. Thus, the mask is a binary matrix of the same size as X. The primary distinction from the original WGAN-GP training algorithm lies in multiplying the generator's output by mask Y, which then serves as the input for discriminator D, thereby generating a score for fake samples. Figure 1 provides an illustration of the proposed training method, showcasing the training routine for both the generator and the discriminator.

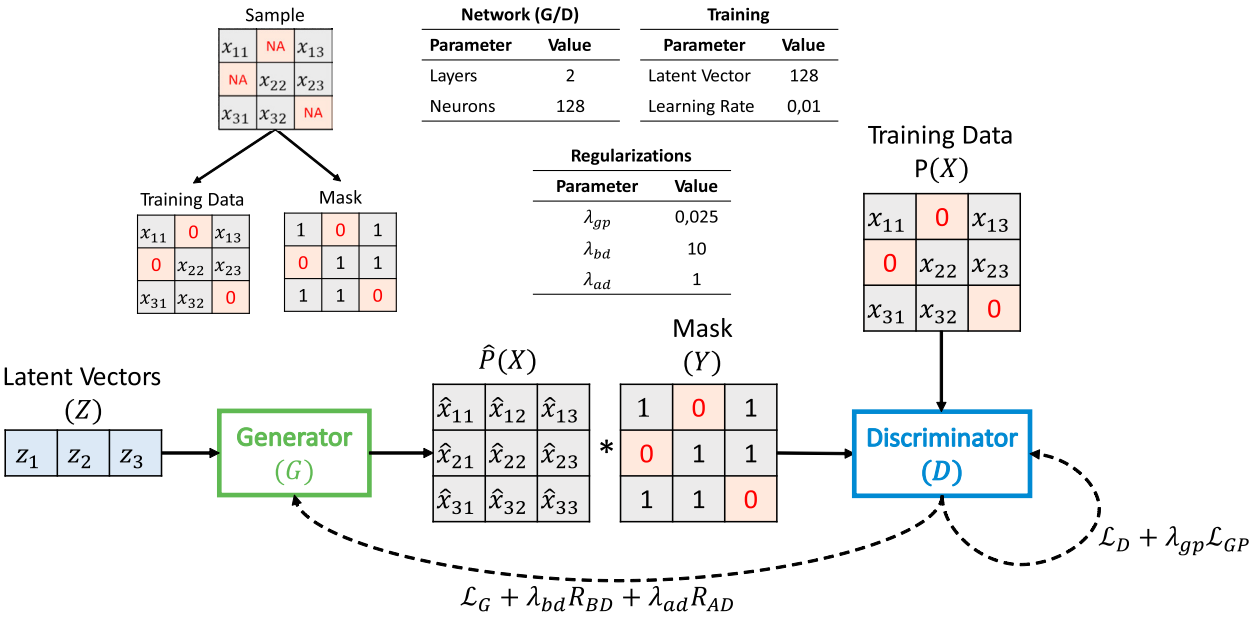


Fig. 1: Illustration of proposed methodology for WGAN-GP models.

3. Case Study

3.1. Travel Survey

The Swedish Institute for Transport and Communications Analysis (SIKA), through the Riks-RVU 2005–2006 national travel behavior survey, provided the data for the study. The survey was conducted from October 2005 to September 2006 and included 41,000 Swedish total population registry stratified samples aged 6–84. Each survey

row is a weighted sample post-stratified by year, area, age, and sex. The regional division is county-level, except for Stockholm County, which is municipal. Age groups are 6–14, 15–24, 25–44, 45–64, and 65–84. For more information about the survey, refer to report by Abramowski and Holmström (2007).

3.1.1. Population Data

We require ground-truth population data without missing information to test the suggested methodology's ability to train WGAN-GP models with missing information. Utilizing the survey, we generate a complete population dataset by reproducing each row's weight value. We removed all rows with missing data from the whole population dataset and kept 17 individual and household features. Table 1 lists the travel survey dataset attributes used in the research, including 13 for individuals and 4 for households. We designate this dataset as *h-population*, which serves as the reference dataset or "ground truth" for subsequent analysis.

Table 1: List of Attributes in the study

SN	Attributes(dim)	Definition	SN	Attributes(dim)	Definition
1	AGE(5)	Age	10	SKOL(2)	Student?
2	SEX(2)	Sex	11	APLATS(2)	Employed?
3	INKUP(5)	Individual income	12	BILANT(8)	Number of cars
4	UP_FORV(3)	Main occupation	13	KKORT_HH(5)	N driving licenses in the household
5	LIVSKAT(8)	Life category	14	HHINK(5)	Household income
6	SUN2KN(7)	Education level	15	HHSTORL(6)	Household size
7	BOST_LAN(21)	Residence County	16	SNIKOD(13)	Industry
8	KORKORT(2)	Driving license possession	17	UP_ANST(3)	Type of employment
9	SUN2KI(10)	Educational focus			

3.1.2. Training Data

For training the proposed WGAN-GP models, we create two different types of training datasets. First, we create a training set that is complete. We sample 10% of the *h-population* and deliberately remove certain unique category combinations from it. We label this refined dataset as *h-nomis*. By eliminating certain combinations in *h-nomis*, we address potential biases in structural zero and sampling zero rates arising from differences in the distributions. Subsequently, we create two incomplete datasets. These datasets are derived from *h-nomis* but include some attributes with missing information. We randomly selected q numbers of attributes and introduced NaN values randomly to $r\%$ of each of the q attributes in the *h-nomis* dataset. We denote these datasets as *h-miss-q-r*, which indicates the number of missing attributes and the proportion of rows with NaN values.

In the study, the WGAN-GP model trained with *h-nomis* acts as the "benchmark" model against which the WGAN-GP models trained on *h-miss-q-r* are analyzed and examined. In the end, we generate a synthetic population from each of the trained models and test them against the *h-population* to evaluate the performance of these models. Table 2 presents a comprehensive summary of the training data employed for the analysis.

Table 2: Statistics on the datasets used for training and analysis.

Dataset	N Rows	N Unique Combinations	Missing Attributes	% of Missing Value per Attribute	% of Missing Rows
h-population	5156896	14811	-	-	-
h-nomis	477489	12811	-	-	-
h-miss-2-30	477489	12803	UP_FORV, SUN2KN	30%	50,98%
h-miss-3-40	477489	12613	UP_FORV, SUN2KN, SUN2KI	40%	78,44%

4. Model Evaluation and Discussion

To validate the effectiveness of our proposed approach for training WGAN-GP with incomplete data, we initially utilized the *h-nomis* dataset to optimize the hyperparameters of the WGAN-GP model. Subsequently, we applied the same optimized parameters to train other models with *h-miss-q-r* datasets. The models underwent training on a GPU cluster comprising four NVIDIA GeForce RTX 3080 units, each with 10GB of memory, for a total of 1000 epochs. Figure 1 details the optimized model parameters.

Ultimately, the best-performing models were evaluated against the *h-population* dataset. To assess the models, we generated synthetic populations named G_{nomis} and $G_{miss-q-r}$ using the model trained on the *h-nomis* and *h-miss-q-r* datasets, respectively. All these synthetically generated populations consist of 5,156,896 data points.

4.0.1. Attribute-level Evaluation

We first perform a column-level check on the WGAN-GP-generated synthetic populations to make sure that each attribute individually is able to follow the statistical distribution of ground-truth data, *h-population*. We perform the attribute level evaluation using three metrics provided by the Datacebo (2024) library: category coverage (score_cc), TV complement (score_tv), and category adherence (score_ca).

Table 3 shows the metric scores for all attributes in the synthetic data generated by different WGAN-GP models, tested against the *h-population* dataset. The presented metric score shows that the proposed WGAN-GP training method successfully trains a model with incomplete data, closely approximating the performance of the benchmark model. Upon further examination of Table 3, the $G_{miss-q-r}$ population performs well for attributes with missing data (UP_FORV, SUN2KN, and SUN2KI) and has similar metrics to G_{nomis}. A closer examination reveals that all populations exhibit lower score_cc for the "KKORT_HH" and "BILANT" attributes. However, such performance levels are deemed acceptable for these trained WGAN-GP models, considering the negligible share of categories that are not generated by models in the actual population.

Table 3: Attribute level evaluation of all WGAN-GP models against h-population

	G_nomis			G_miss-2-30			G_miss-3-40		
Attributes	score_cc	score_tv	score_ca	score_cc	score_tv	score_ca	score_cc	score_tv	score_ca
AGE	1,0	0,99	1,0	1,0	0,98	1,0	1,0	0,98	1,0
SEX	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
INKUP	1,0	0,99	1,0	1,0	0,99	1,0	1,0	1,0	1,0
UP_FORV	1,0	0,99	1,0	1,0	0,99	1,0	1,0	0,99	1,0
LIVSKAT	1,0	0,99	1,0	1,0	0,98	1,0	1,0	0,98	1,0
KORKORT	1,0	0,99	1,0	1,0	1,0	1,0	1,0	0,99	1,0
UP_ANST	1,0	1,0	1,0	1,0	0,99	1,0	1,0	1,0	1,0
SUN2KN	1,0	0,98	1,0	1,0	0,99	1,0	1,0	0,94	1,0
SUN2KI	1,0	0,99	1,0	1,0	0,98	1,0	1,0	0,95	1,0
SKOL	1,0	1,0	1,0	1,0	0,99	1,0	1,0	0,99	1,0
APLATS	1,0	1,0	1,0	1,0	0,99	1,0	1,0	1,0	1,0
BOST_LAN	1,0	0,97	1,0	1,0	0,94	1,0	1,0	0,95	1,0
BILANT	0,88	0,98	1,0	0,88	0,99	1,0	0,75	0,98	1,0
KKORT_HH	0,80	0,99	1,0	0,80	0,98	1,0	0,80	0,98	1,0
HHINK	1,0	0,98	1,0	1,0	0,99	1,0	1,0	0,99	1,0
HHSTORL	1,0	0,98	1,0	1,0	0,98	1,0	1,0	0,98	1,0
SNIKOD	0,92	0,97	1,0	0,92	0,98	1,0	0,92	0,98	1,0
Average	**0,976**	**0,986**	**1,0**	**0,976**	**0,985**	**1,0**	**0,969**	**0,982**	**1,0**

4.0.2. Higher Dimension Evaluation

Following Garrido et al. (2020) and Borysov et al. (2019), we compare the categorical partial joint distribution of synthetic populations generated by WGAN-GP models. We used a standardized root mean square error (SRMSE) and coefficient of determination R^2 as metrics for evaluating multi-dimensional distributions. We employ distinct subsets

of attributes to encompass diverse features of the joint distribution. The evaluation is done for the following k-joint of attributes, with total categories in brackets:

- 210-dimensional joint of AGE(5), SEX(2) and BOST_LAN(21).
- 16k-dimensional joint of UP_FORV(3), SUN2KN(7), SUN2KI(10), HHSTORL(6) and SNIKOD(13).
- 7M-dimensional joint of AGE(5), SEX(2), BOST_LAN(21), SUN2KI(10), SNIKOD(13), LIVSKAT(8), SUN2KN(7), and INKUP(5).

Table 4 shows the results from the SRMSE and R^2 metrics for all WGAN-GP models at different k-joint levels. It also displays the total number of unique category combinations generated for each k-level (Total), the number of actual combinations in the *h-population* (Real), and the number of combinations present in each synthetic population for all k-level joints.

The findings indicate that for all $G_{miss-q-r}$ populations, their performance falls short of the benchmark G_{nomis} population in terms of both SRMSE and R^2 values across all k-dimensional joints. Consequently, as the amount of missing information in the training data increases, the SRMSE values tend to rise (and corresponding R^2 values falls), with the $G_{miss-3-40}$ population exhibiting the highest error. In general, the $G_{miss-q-r}$ population exhibits SRMSE and R^2, are within acceptable range of error from benchmark population G_{nomis}. Therefore, we can conclude that the proposed WGAN training method successfully trains a model with incomplete data.

It is noteworthy that with an increase in the number of joint combinations, the SRMSE value rises (while R^2 decreases) across all models, as seen in 16k and 7M dimensional joints. This phenomenon occurs because the WGAN models generate a substantially larger number of category combinations compared to the ground-truth population. We analyze this behavior by examining the count of sampling and structural zero data points produced by the models and evaluating the models' performance concerning the category combinations.

Table 4: Higher dimension evaluation for k-joint level distribution of attributes for all models

	210-dimensional (Total:210, Real:210)			16k-dimensional (Total:16380, Real:2485)			7M-dimensional (Total:7644000, Real:10395)		
Model	N Comb. Generated	SRMSE	R2	N Comb. Generated	SRMSE	R2	N Comb. Generated	SRMSE	R2
G_nomis	210	0,19	0,981	11897	1,75	0,978	365530	25,79	0,913
G_miss-2-30	210	0,21	0,976	11372	1,97	0,973	349566	26,61	0,907
G_miss-3-40	210	0,25	0,967	12022	2,26	0,964	414874	26,82	0,906

4.0.3. Sampling and Structural Zero Evaluation

Based on the definitions provided in Kim and Bansal (2023), we extracted general sample, sampling zero, and structural zero data for each generated population. We carried out this process for the k-joint attribute combination, as detailed in Section 4.0.2, and at various data sampling levels. Figure 2 illustrates the different type of data types and the region of overlaps across different population distribution for calculating general score, sampling rating and structural score. We computed the ratio of both the general sample and sampling zero in the generated population in comparison to the combinations present in the *h-population*. We determine the ratio of structural zeros by calculating the total number of structural zero instances generated by the WGAN-GP models against all the unique combinations produced by the model itself.

Figure 3 displays the evaluation plots for all models, obtained from synthetic data generated for 16k-dimensional and 7M-dimensional categorical joints. Both plots demonstrate that all WGAN-GP models have similar metrics and are close to the benchmark G_{nomis} population. The metric score shows that the proposed WGAN-GP training approach trains a model with incomplete data that closely matches the benchmark model. Deeper analysis of the results shows that all models generated virtually all ground-truth population general samples and sampling zeros as the sampling rate increased. For instance, a 7M-dimensional distribution with 5M data points captures 93.69% $\pm$ 0.36% of general samples and 89.42% $\pm$ 0.8% of sampling zeros. Furthermore, as the sampling rate and k-dimensional space grow, all

models have more structural zeros. This is because additional attribute dimensions allow for more data combinations. The *h-population* in this study contains only 10 395 (0.13%) unique combinations out of 7M potential possible combinations. The WGAN-GP model can't stop structural zero generation for such a small set of unique combinations of ground-truth data, especially for attribute combinations in high-dimensions.

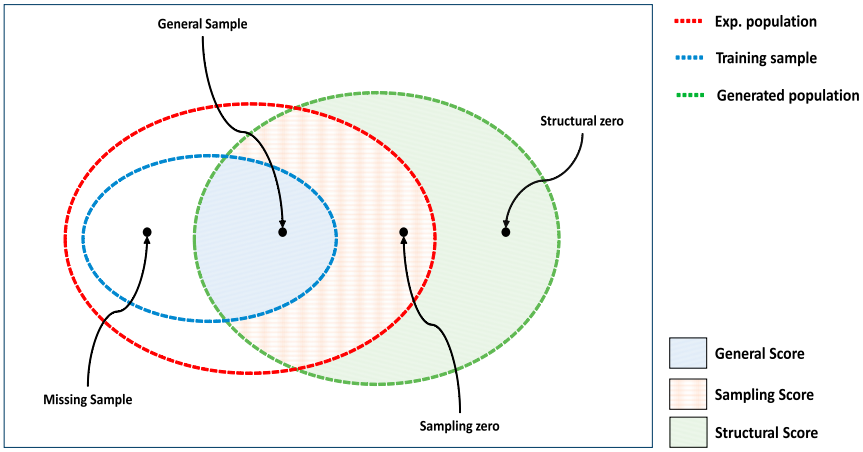
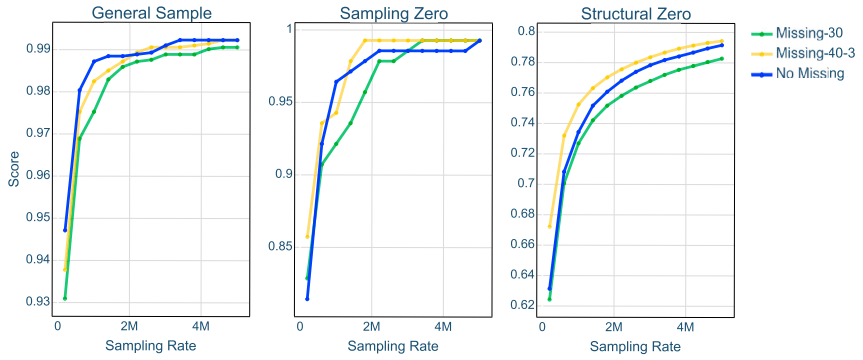
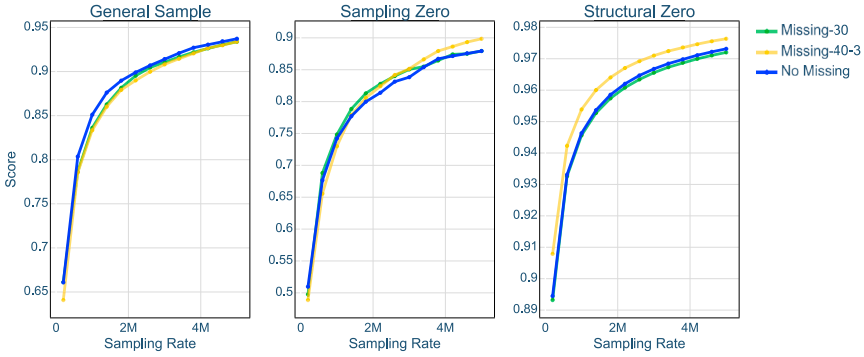


Fig. 2: Conceptual diagram showing the distribution of combination of categories and data types for our study.



(a) 16k-dim joint



(b) 7M-dim joint

Fig. 3: Plots with the ratio of general sample, sampling zero, structural zero for 16k and 7M dimensional joint data at different sampling levels.

5. Conclusion

The paper introduces a novel method for population synthesis using WGAN-GP models, which effectively train on incomplete data to guarantee the completeness of the synthetic population the models generate. This property of the model is especially useful when publicly accessible microsamples have missing information on one or more attributes due to errors in data collection, privacy concerns resulting in data being withheld, or when information is missing during the merging of multiple microsamples. The proposed methodology utilizes a mask matrix to depict missing values in training data, which allows the WGAN-GP model to train on datasets that contain missing attributes.

The training method was validated using data from the Swedish national travel survey. We conducted a comparison between the benchmark model, trained on complete data, and models, trained on data with varying levels of missing information. The population generated from all the trained models was evaluated at the attribute-level and higher k-dimensional levels to assess the model's capability in generating sampling and structural zeros. For all the evaluation metrics, the results obtained from the trained WGAN-GP on incomplete data exhibit a high degree of similarity with the benchmark WGAN-GP model trained on complete data. The validation results affirm the efficacy of the suggested training technique employing a mask matrix in proficiently managing incomplete data, leading to synthetic populations that closely resemble the real population.

The paper makes a substantial contribution to the field by providing a strong solution for population synthesis using incomplete data. The new finding opens up new areas for research in the future and shows how deep generative models can improve population synthesis, which is important for ABMs used in transportation simulations and other areas.

5.1. Limitations

The analysis conducted in this study is restricted to just two datasets with missing values, specifically *h-miss-2-30* and *h-miss-3-30*. Consequently, the findings offer only a narrow insight into the potential of the proposed method. To gain a more comprehensive understanding of its capabilities to learn from missing data, future research will explore a wider array of *h-miss-q-r* datasets, allowing for a more extensive evaluation.

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